

LLM Model Comparison Guide

Efficiency, functionality, and best-use recommendations across major model families — August 2026

A note on currency: a few names have moved on since compiled. GPT went past 5.2 to 5.4 and 5.5; Gemini has 3.1 Pro and newer 3.5 Flash; Claude's current lineup (Sonnet 5, Opus 4.8, Haiku 4.5) supersedes Opus 4.6 / Sonnet 4.6. This guide covers what was requested and flags where things have shifted.

DeepSeek

Model	Efficiency	Functionality	Best For
DeepSeek R1	Reasoning model, slower per query (chain-of-thought); open-weight and cheap to self-host	Strong math/logic/code reasoning, competitive with early o1-class models	Budget-conscious reasoning tasks, research, self-hosted agentic pipelines
DeepSeek V3 0324	Fast, non-reasoning, very cheap via API	Solid general chat/coding; MoE architecture keeps inference cost low	High-volume chat apps, cost-sensitive production use

Meta Llama

Model	Efficiency	Functionality	Best For
Llama 3.2 90B Vision Instruct	Moderate — vision adds compute cost	Multimodal (image + text) instruction following	Document/image Q&A, OCR-adjacent tasks, open-source multimodal apps
Llama 3.3 70B Instruct	Efficient for its capability tier	Strong general-purpose text tasks, comparable to older 405B model	Self-hosted assistants, fine-tuning base, on-prem deployments
Llama 4 Maverick 17B-128E	MoE — 17B active params, cheap inference despite huge total param count	Long context, decent multimodal, good throughput	High-throughput open deployments needing big effective capacity without full dense-model cost

Microsoft Phi

Phi 4 Reasoning — small (14B-class) model tuned specifically for step-by-step reasoning via distillation from larger models. Very cheap to run, even on-device. **Best for:** edge/local deployment, math/logic tasks where you can't afford a frontier model's cost or latency.

OpenAI

Model	Efficiency	Functionality	Best For
GPT-4.1 / Mini / Nano	Nano is extremely cheap/fast; 4.1 is mid-tier	Solid general tasks; Nano trades quality for speed/cost	Nano: simple classification/extraction at scale; 4.1: general apps
GPT-4o / 4o Mini	4o Mini very cheap	Native multimodal (voice, vision), older but stable	Voice apps, legacy integrations

Model	Efficiency	Functionality	Best For
GPT-3.5 Turbo	Very cheap, fast	Dated capability, mostly legacy	Legacy compatibility only — not recommended for new builds
GPT-5 Mini	Cheap, fast reasoning-lite	Good balance for a smaller reasoning model	Cost-sensitive apps needing better-than-4o quality
GPT-5.0 / 5.1	Reasoning-capable, moderate cost	Strong general + coding, "Thinking" modes	General knowledge work
GPT-5.2	Family with instant, thinking, and Pro modes; Pro uses more compute than thinking	Strong reasoning tier of its generation	Complex analysis, agentic tasks (now succeeded by 5.4/5.5)
GPT-5.2 Codex / 5.3 Codex	Coding-specialized; 5.3-Codex reported ~25% faster than Claude Opus 4.6	Agentic coding, repo search, terminal/tool use, debugging	Autonomous coding agents, large-repo refactors
GPT-5.4	Most capable/efficient frontier model at release; 1M-token context, tool search, better token efficiency than 5.2	Coding, computer use, long-context agentic work	Enterprise agent workflows needing huge context + efficiency
GPT-5.4 Mini / Nano	Cheap tier of 5.4 line (though ~4x pricier via API than GPT-5 equivalents)	Same family capabilities, scaled down	Cost-sensitive high-volume 5.4-quality tasks

Since this list was made, OpenAI shipped **GPT-5.5** — its first fully retrained base model since 4.5, optimized for agent-oriented objectives at the pretraining level rather than just fine-tuned afterward. It's currently their top agentic/coding model.

Google Gemini

Model	Efficiency	Functionality	Best For
Gemini 2.5 Flash / Flash-Lite	Very cheap, fast	Good everyday multimodal tasks	High-volume, latency-sensitive apps
Gemini 2.5 Pro	Moderate cost	Strong reasoning + huge context	Long-document analysis
Gemini 3 Flash Preview	Fast, \$0.50/M input & \$3.00/M output, 1M-token context	Text, image, audio, video, tool use, structured data	Cost-efficient agentic/multimodal apps
Gemini 3.1 Pro Preview	Higher cost, slower, but highest reasoning ceiling	Frontier reasoning across text/audio/image/video/PDFs/code repos, 1M-token context	Hardest reasoning/math tasks where ceiling matters more than cost

Google has since released **Gemini 3.5 Flash** and is expected to move toward a 3.5 Pro — worth checking if you have access, as 3.5 Flash already beats 3.1 Pro on several agent/coding benchmarks at lower cost.

Claude (Anthropic) — as listed

Haiku 4.5: fast, cheap, strong for its size — good for high-volume, latency-sensitive tasks (chat support, quick edits, simple agents).

Opus 4.6 / Sonnet 4.6: strong general + coding models in their generation, but Anthropic's current lineup is **Sonnet 5**, **Opus 4.8**, and **Haiku 4.5** — the 4.6 models have been superseded. Prefer the current versions for the same "grade" (e.g., Opus 4.8 instead of Opus 4.6).

Quick "Best For" Cheat Sheet

Cheapest/fastest for simple tasks: GPT-4.1 Nano, Llama 3.3 70B, Gemini 2.5 Flash-Lite, Haiku 4.5, DeepSeek V3

Best raw reasoning/math: Gemini 3.1 Pro, GPT-5.5/5.4, DeepSeek R1, Phi 4 Reasoning (small-model constraint)

Best agentic coding: GPT-5.3/5.4 Codex line, Claude Opus-tier models, Gemini 3.1 Pro

Best multimodal (vision/audio/video): Gemini 3 series, Llama 3.2 90B Vision, GPT-4o

Best for self-hosting / open-weight control: DeepSeek (R1, V3), Llama 4 Maverick, Llama 3.3 70B, Phi 4 Reasoning

Best long-context (1M+ tokens): GPT-5.4, Gemini 3 series

Tip: For a tailored recommendation, share your actual use case (coding agent, customer support bot, research assistant, on-device app, budget constraints, etc.) and this can be narrowed to 2–3 realistic picks.